Stanley Nelavala

Backend and Platform Software Engineer
218-213-8220 | consultantstanleymn@gmail.com | LinkedIn | GitHub | stanley-n.com
Open to relocation: SF Bay Area, New York, Seattle

Summary

Software engineer with an M.S. in Computer Science (4.0 GPA) and one year on a 10-person hospital operations platform. I build Python and FastAPI services, React and React Native apps, and GitHub Actions and AWS release pipelines. Results: a catalog query cut from 285 ms to 0.20 ms, a regex denial-of-service bug closed, mobile test coverage raised from 14% to 93%. I also build AI-agent and reliability projects with public code and tests. AWS Certified Solutions Architect, Associate.

Experience

Software Development Engineer Nov 2025 – Present
Mass Technologies, team of 10 (RxApps360 hospital platform, pre-launch; all numbers from dev and test environments) *Huntington, IN*

- Replaced offset paging with keyset paging on the catalog device list: on a 328,000-device dev database, the query for row 150,000 fell from 285 ms to 0.20 ms. Cut the planning Items page from 3 API calls and 162 KB to 2 calls and 37 KB with typeahead search and cursor paging.
- Designed, built and integrated into every platform module the signed service-to-service API (M2M): each call proves which service and tenant sent it. After a security audit, hardened it with a Redis rate limiter and replay protection.
- Closed a regular expression denial-of-service bug in a shared validation package: an 80-character hostile input took about 138 s; the new linear pattern matched the old results on 60,000 random inputs.
- Built the catalog search API on a shared service layer with keyset cursors and a unique-id tie-break, so pages never skip or repeat rows; tested with rows inserted, deleted and tied between pages.
- Built RxDesign, the hospital design service, backend and web (FastAPI, React, TypeScript), with a deterministic, rule-based layout engine that places rooms on a 0.5 ft grid with doors, exits and flow lines; 220+ offline tests on the API.
- Built the RxAsset iOS and Android app (React Native, Expo) with scan-to-search and an on-device voice assistant (no cloud call); set up separate iOS and Android GitHub Actions release pipelines with TestFlight submission; raised statement coverage from 14% to 93% (582 tests in 68 suites).
- Standardized malware scanning of file uploads (GuardDuty) across 9 app modules; load-tested sign-in with JMeter at 100 concurrent users; automated weekday start and stop of dev and acceptance servers and databases (ECS, RDS), so they run about 53 hours a week instead of 168.

Adjunct Faculty (part-time) Jan 2026 – Aug 2026
The College of St. Scholastica *Duluth, MN*

- Taught Programming II (Java) and Machine Learning for Data Science.

Graduate Research Assistant and Teaching Assistant Aug 2023 – Aug 2025
University of Minnesota Duluth *Duluth, MN*

- Built a pipeline (Python, Node.js, MySQL, FFmpeg, PaddleOCR) that collected and analyzed 1M+ TikTok videos from 300k+ authors for thesis research on misinformation, then grouped themes with BERTopic; presented at the UMD CS graduate poster session. Led labs and office hours for Python, NLP and Parallel Programming.

Projects (public code, tests)

Agent Integration Platform | *Python, FastAPI, LangGraph, MCP, PostgreSQL (pgvector), Redis, Terraform* GitHub

- LangGraph case workflow with typed state, a Postgres checkpointer, human approval that resumes after restart, and step and token budgets; 62 deterministic replay evals, prompt-injection tests, PII redaction and an append-only audit log.
- Hybrid retrieval (pgvector and full-text search fused by RRF) with access labels; failover between two LLM provider adapters with a circuit breaker and token cost records.

Payments Ledger Service | *Java 25, Spring Boot, PostgreSQL, Flyway, Kafka API, Prometheus, Grafana, k6* GitHub

- Three user-facing SLIs with 28-day error budgets and burn-rate alerts tested with promtool; a scripted failure lab (Kafka down, Postgres restart, poison event, latency injection) with balance checks after each.
- Rolled forward and back by image digest across an expand-and-contract Flyway migration; k6 SLO thresholds as a gate.

Skills

Languages: Python, TypeScript, JavaScript, SQL, Java, Go, Rust
Backend and data: FastAPI, Spring Boot, REST, gRPC, PostgreSQL, Redis, Kafka API, migrations (Alembic, Flyway)
AI and agents: LangGraph, MCP, hybrid retrieval and RAG, deterministic LLM evals, guardrails
Frontend, mobile, cloud: React, Next.js, React Native, Expo, AWS (ECS, RDS, S3, SQS), Terraform, Docker, GitHub Actions
Testing and reliability: pytest, Jest, Playwright, k6, OpenTelemetry, Prometheus, Grafana, SLOs

Education and Certification

M.S. Computer Science, University of Minnesota Duluth, 2023 – 2025, GPA 4.0, Graduate Summer Research Fellowship 2024 and 2025
B.Tech Computer Science and Engineering, IIIT D&M Kancheepuram, 2019 – 2023
AWS Certified Solutions Architect, Associate, Feb 2025 – Feb 2028 (verifiable on Credly)